NASIF IMTIAZ

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EDUCATION

North Carolina State University (NCSU), Raleigh, NC **Aug 2017 - May 2023 (Expected)**
Ph.D. in Computer Science
- Dissertation title: Toward Secure Use of Open Source Dependencies.

North Carolina State University (NCSU), Raleigh, NC **2022**
M.S. in Computer Science

Bangladesh University of Engineering and Technology (BUET), Dhaka, Bangladesh **2017**
B.Sc. in Computer Science and Engineering

EXPERIENCE

Graduate Research Assistant **August 2017 - Present**
North Carolina State University
- Research in the area of Software Engineering & Software Security.

Software Engineer Intern **Summer 2021, 2022**
Systems & Infra *Meta*
- Built support so that Rust services can use C++ infra through Rust/C++ asynchronous interop.
- Developed server-side streaming support in Rust for Thrift RPC framework.
- Developed a Rust dependency analysis tool, depdive.

Research Intern **Summer 2019**
One Engineering Systems *Microsoft Research*
- Built supervised models to predict and explain developer actions on security warnings.
- Conducted interviews and surveys to collect feedback and identify features.

TECHNICAL SKILLS

- **Languages:** Rust, Python, C/C++, Java, SQL, R.
- **Research Skills:** Mixed-methods, Data mining, Statistical modeling, Machine Learning.
- **Tools/Systems:** Linux/Unix, Git, Pandas, NumPy, Scikit-learn, Vue.js.

SELECTED PUBLICATIONS

Open or Sneaky? Fast or Slow? Light or Heavy?: Investigating Security Releases of Open Source Packages **2022**
- IEEE Transactions on Software Engineering

A comparative study of vulnerability reporting by software composition analysis tools **2021**
- 15th ACM/IEEE International Symposium on Empirical Software Engineering and Measurement (*acceptance rate: 19.4%*)

Investigating the Effects of Gender Bias on GitHub **2019**
- 41st ACM/IEEE International Conference on Software Engineering (*acceptance rate: 21%*)

PROJECTS

Supply Chain Security **Aug 2019 - Present**
Research Project *NCSU*
- Building a collaboration graph to provide trust ratings for open source developers through social network analysis [*ongoing*].
- Built and evaluated tooling to analyze reproducibility and code review coverage of package updates.

Developer Adoption of Security Tools **Aug 2018 - Aug 2020**
Research Project *NCSU*
- Analyzed the performance and developer adoption of security tools that perform static analysis, credential scanning, and software composition analysis. Collaborated with Microsoft and RunSafe Security.

GRADUATE COURSES

Automated Learning and Data Analysis, Foundations of Data Science, Artificial Intelligence, Software Engineering, Compiler Construction, Digital Logic Design, Software Testing, Software Engineering as Human Activity, User Experience.

SELECTED GRADUATE COURSE PROJECTS

Object Classification on Caltech-256 Dataset
- Used multiple deep learning architectures and optimization techniques to classify objects in images.

Prototxt to TensorFlow converter
- Built compiler to generate python TensorFlow code from neural networks defined in prototxt file.